Phase Response fn Rational Transfer function.

$$H(z) = \frac{b_0 \prod_{k=1}^{M} (1 - q_k z^{-1})}{\prod_{k=1}^{N} (1 - p_k z^{-1})}$$

q_k are zeros

p_k are poles

$$H(\omega) = H(z)\Big|_{z = e^{j\omega}}$$

$$= \frac{b_0 \prod_{k=1}^{M} (1 - q_k e^{-j\omega})}{\prod_{k=1}^{N} (1 - p_k e^{-j\omega})}$$

$$\Phi(\omega) = \angle H(\omega) = \arg(H(\omega))$$

$$= \arg(b_0) + \text{Effect of zeros} \rightarrow + \sum_{k=1}^{M} \arg(1 - q_k e^{-j\omega}) - \text{Effect of poles} \rightarrow - \sum_{k=1}^{N} \arg(1 - p_k e^{-j\omega})$$

Generic term $\arg(1 - re^{j\theta} e^{-j\omega})$

$$= \arg(1 - r\cos(\omega - \theta) + jr\sin(\omega - \theta))$$

$$= \tan^{-1}\left(\frac{r\sin(\omega - \theta)}{1 - r\cos(\omega - \theta)}\right)$$

Geometric interpretation for a single zero (real)

$$H(z) = 1 - rz^{-1}$$

⇒ Real impulse response.

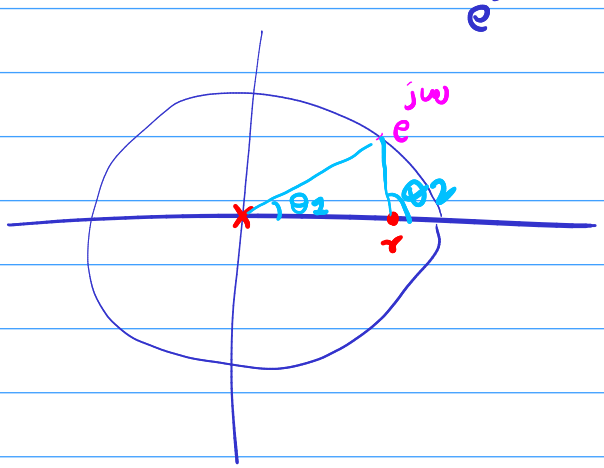
Phase response is odd function

$$H(\omega) = 1 - re^{-j\omega} = \frac{e^{j\omega} - r}{e^{j\omega}}$$

$$\boxed{\Phi(\omega) = -\Phi(-\omega)}$$

$$\arg H(\omega) = \arg(e^{j\omega} - r) - \arg(e^{j\omega})$$

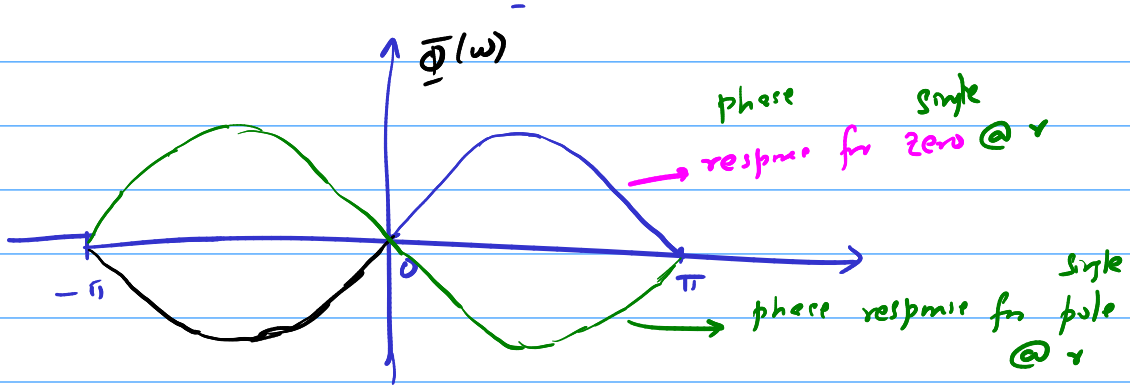
$$(\theta_2) \qquad (\theta_1)$$



$$\phi(\omega) = \theta_2 - \theta_1$$

$\omega = 0 \quad \theta_1 = \theta_2 = 0 \quad \Phi(\omega) = 0$

$\omega = \pi \quad \theta_1 = \theta_2 = \pi \quad \Phi(\omega) = 0$

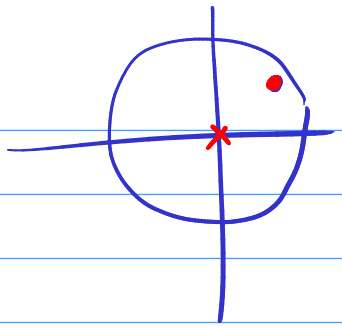


- Phase response of single complex pole

$$H(z) = \frac{1}{1 - rz^{-1}}$$

$\Phi(\omega)$ = − response of single zero

Single complex zero

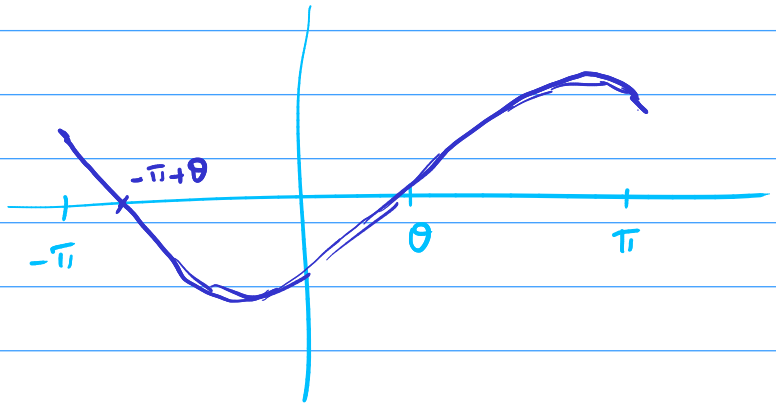


$$H(z) = 1 - re^{j\theta} z^{-1}$$

$$H(\omega) = 1 - re^{-j(\omega-\theta)}$$

$\Downarrow$

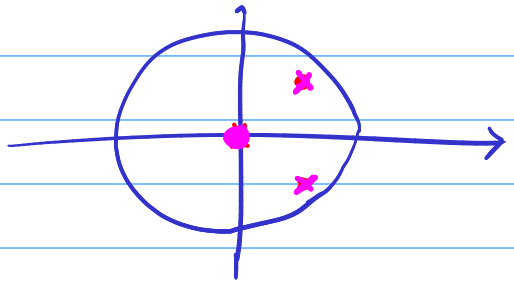
Phase response gets shifted by θ.



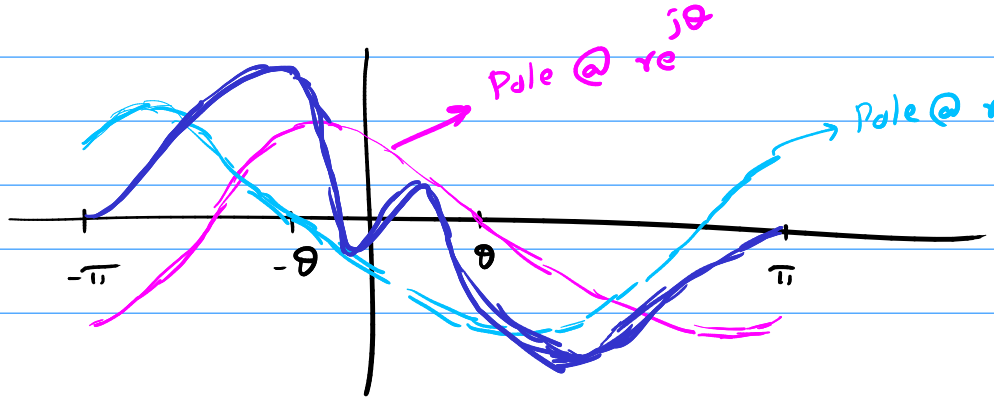
- Second order. Poles (resonator)

$$H(z) = \frac{1}{(1 - re^{j\theta} z^{-1})(1 - re^{-j\theta} z^{-1})}$$

(real valued impulse response)



$\Phi(\omega)$ = response for Pole @ $re^{j\theta}$ + resp for Pole $-\theta$ @ re

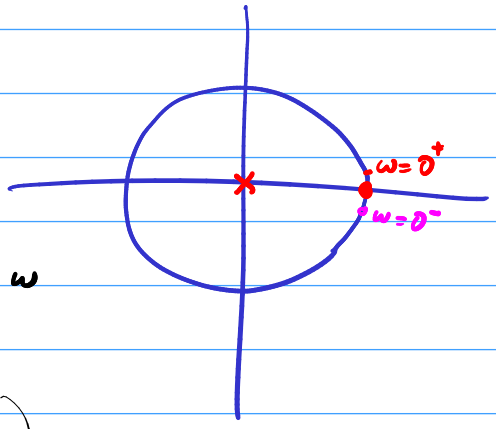


Zeros on unit circle (Important case)

$$H(z) = 1 - z^{-1}$$

$$H(\omega) = 1 - e^{-j\omega}$$

$$= 1 - \cos\omega + j \sin\omega$$

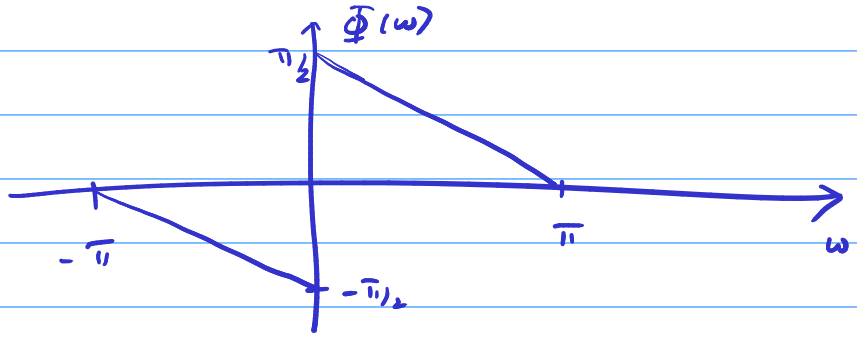


$$\Phi(\omega) = \tan^{-1}\left(\frac{\sin\omega}{1-\cos\omega}\right)$$

$\omega > 0$ $\quad \tan\left(\pi/2 - \omega/2\right) = \dfrac{\sin\omega}{1-\cos\omega}$

$$\Phi(\omega) = \begin{cases} \pi/2 - \omega/2 & ; \ \omega \geq 0 \\ -\pi/2 - \omega/2 & ; \ \omega < 0 \end{cases}$$

odd function



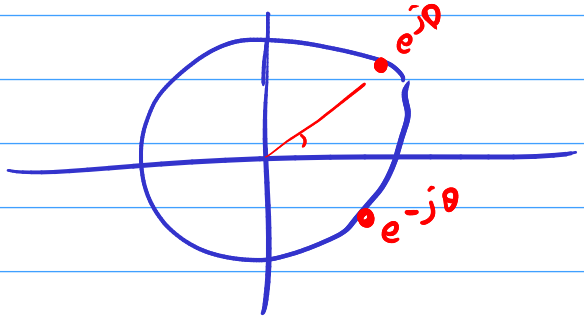
When zero is on unit circle

1. Phase response is linear in ω
2) Slope of line $-1/2$
3) Phase jump of π at $\omega = 0$

(location of zero of transfer funct.)

Two zeros on unit circle

$$H(z) = (1 - e^{j\theta} z^{-1})(1 - e^{-j\theta} z^{-1})$$



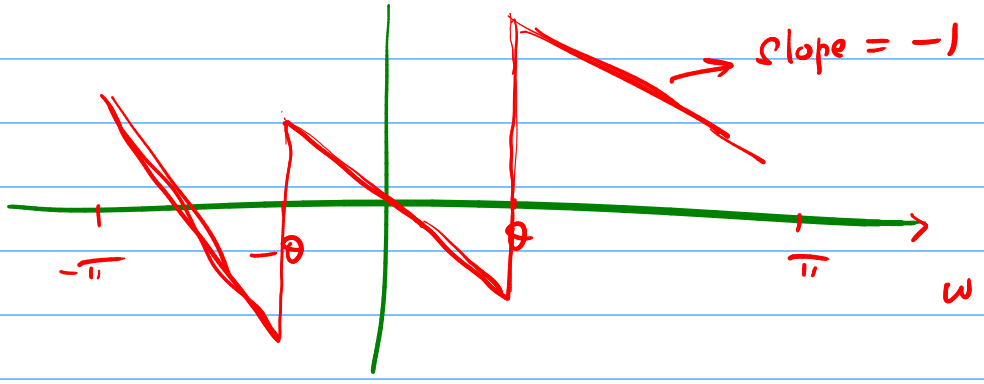
Phase resp $\Phi(\omega)$ = phase resp fn zero @ $e^{j\theta}$ + Phase resp fn zero @ $e^{-j\theta}$

⇓

Straight line slope $-1/2$ — Phase jump π @ $\omega = \theta$

Straight line slope $-1/2$ — Phase jump π at $\omega = -\theta$

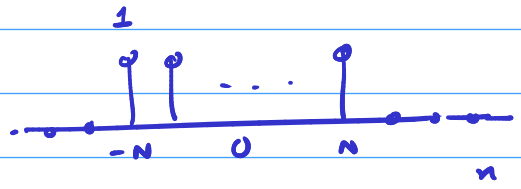
= straight line slope = -1



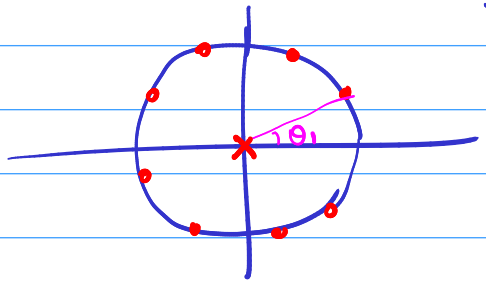
Zeros on Unit circle

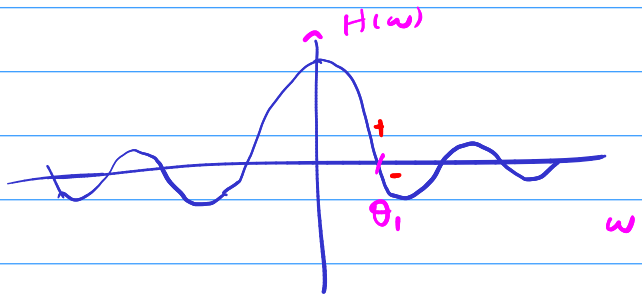
$h(n) = 1 \quad ; \quad -N \le n \le N$

(non causal)

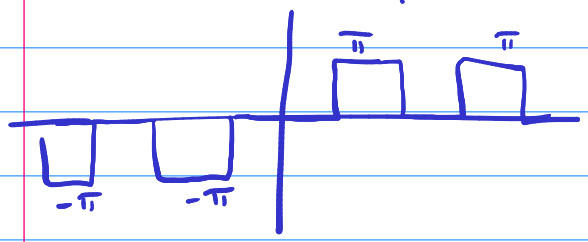


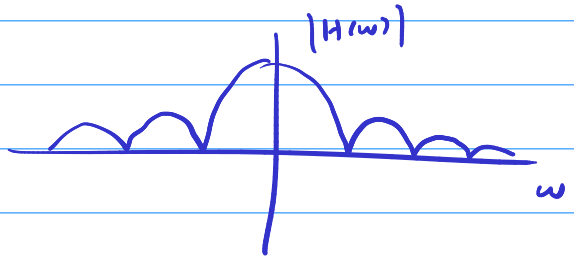
$H(\omega) = \dfrac{\sin\left(\frac{2N+1}{2}\omega\right)}{\sin(\omega/2)}$ (real valued)





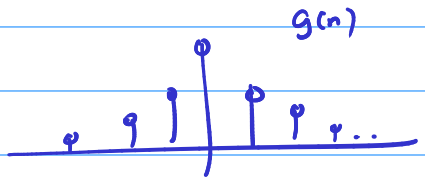
$\angle H(\omega) = \phi(\omega)$





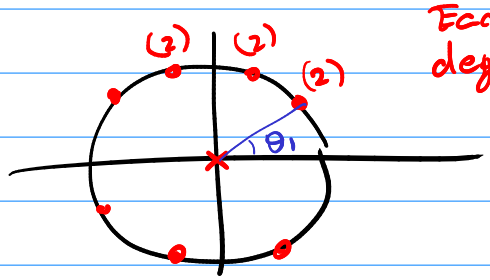
Second order zeros

$\{g(n)\} = \{h(n)\} * \{h(n)\}$



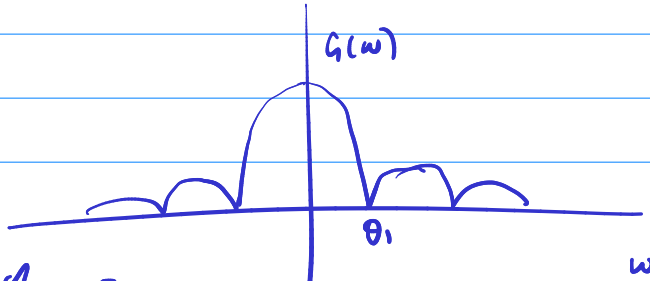
$G(z) = H(z) \cdot H(z) = (H(z))^2$

Each zero has degree 2



$G(\omega) = (H(\omega))^2$

$= \dfrac{\sin^2\left(\frac{2N+1}{2}\omega\right)}{\sin^2\left(\frac{\omega}{2}\right)}$



Phase jump of 2π as ω crosses the location of 2nd order zero

Causal Sequence

$$g(n) = h(n-N)$$

$$= \begin{cases} 1 ; & 0 \le n \le 2N \\ 0 ; & \text{else} \end{cases}$$

$$G(\omega) = H(\omega) \cdot e^{-j\omega N}$$

($e^{-j\omega N}$: linear phase)

$$\theta(\omega) = -\omega N$$

We can write

$$G(\omega) = A(\omega)\, e^{j\theta(\omega)}$$

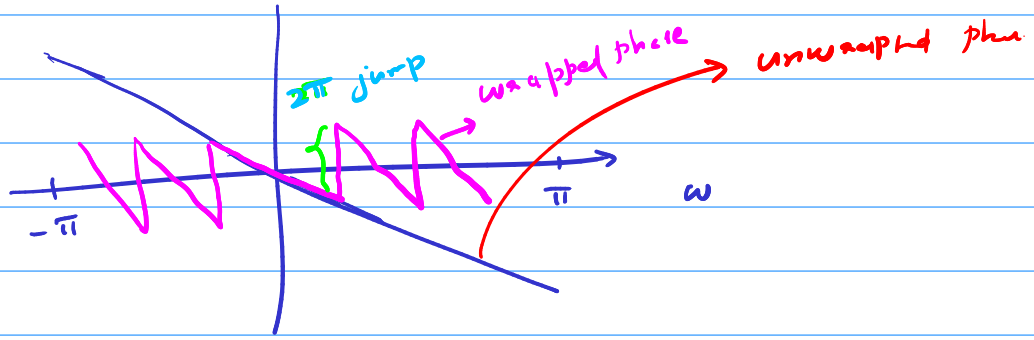
$A(\omega)$ => Real valued function (Amplitude response)

$\theta(\omega)$ => phase response.

In this representation, there is no phase jump of π

Wrapped & Unwrapped phase

$\theta(\omega) = -\omega N$ — Wrapped phase => restrict to $-\pi$ to π



Group Delay Response.

$$H(\omega) = e^{-j\omega N + \theta}$$

$$x(n) = e^{j\omega n} \longrightarrow \boxed{H(\omega)} \longrightarrow y(n) = e^{j\omega(n-N)+\theta}$$

$x(n)$ gets delayed by N samples.

Definition

Group delay response $\tau(\omega) = -\frac{d}{d\omega}\phi(\omega)$

For any LTI system.

$\phi(\omega) \Rightarrow$ Phase response

First order zero

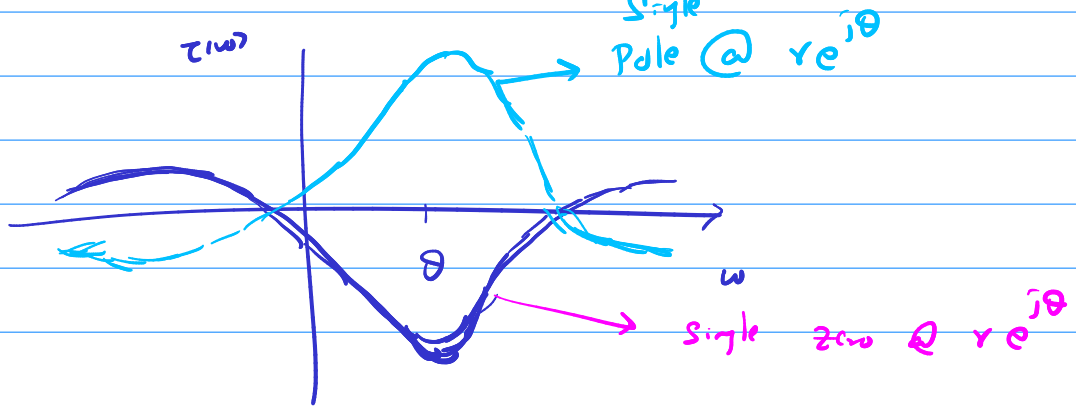
$$H(z) = 1 - re^{j\theta} z^{-1}$$

$$H(\omega) = 1 - re^{j\theta} e^{-j\omega}$$

Phase resp $\phi(\omega) = \arg\left(1 - r\cos(\omega-\theta) + jr\sin(\omega-\theta)\right)$

Group delay

$$\tau(\omega) = -\frac{d}{d}\phi(\omega) = \frac{-d}{d\omega}\tan^{-1}\frac{r\sin\omega-\theta}{1-r\cos\omega-\theta}$$



Zeros on unit circle.

$$H(z) = 1 - z^{-1}$$

$$\phi(\omega) = \begin{cases} \pi/2 - \omega/2 \; ; \; \omega \geq 0 \\ -\pi/2 - \omega/2 \; ; \; \omega < 0 \end{cases}$$

Group delay resp $\tau(\omega) = -\frac{d}{d\omega}\phi(\omega)$

$$= 1/2$$

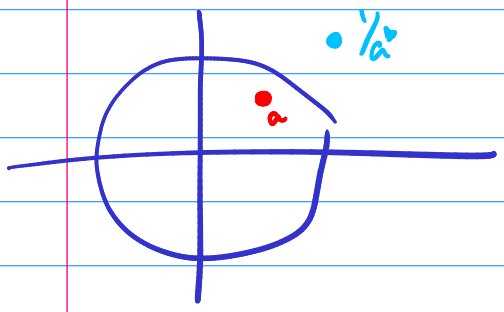
Constant Group delay

All Pass & Minimum Phase Systems

Consider $A(z) = 1 - az^{-1}$ zero @ $z = a = re^{j\theta}$

$B(z) = z^{-1} - a^*$ zero @ $z = 1/a^*$

$= \frac{1}{r} e^{j\theta}$



$A(\omega) = 1 - ae^{-j\omega}$

$B(\omega) = e^{-j\omega} - a^*$

$= e^{-j\omega}(1 - a^* e^{+j\omega})$

$B(\omega) = e^{-j\omega} A^*(\omega)$

$|B(\omega)| = 1 \cdot |A^*(\omega)| = |A(\omega)|$

$\angle B(\omega) \neq \angle A(\omega)$

$C(z) = \frac{1}{1 - az^{-1}}$ $\quad D(z) = \frac{1}{z^{-1} - a^*}$

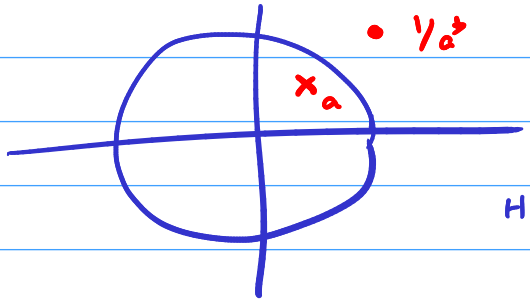
$|C(\omega)| = |D(\omega)|$

Zero/Pole @ a can be replaced with zero/pole @ $1/a^*$ without affecting magnitude response.

1. All Pass syste.

$$H(z) = \frac{z^{-1} - a^*}{1 - a z^{-1}} = \frac{B(z)}{A(z)} \quad |a| < 1$$

Causal syste.



ROC: $|z| > |a|$

$$H(\omega) = \frac{B(\omega)}{A(\omega)}$$

$$|H(\omega)| = \frac{|B(\omega)|}{|A(\omega)|} = 1 \quad \text{for all } \omega.$$

$$\angle H(\omega) = \angle B(\omega) - \angle A(\omega)$$

$$\phi(\omega) \overset{\text{lot of algebra}}{=} -\omega - 2\tan^{-1}\left(\frac{r\sin(\omega-\theta)}{1 - r\cos(\omega-\theta)}\right)$$

Group delay respne

$$\tau(\omega) = -\frac{d\phi(\omega)}{d\omega}$$

$$= \frac{1 - r^2}{|1 - re^{j\theta} e^{-j\omega}|^2}$$

Since For stable & causal syste $r < 1$

$$\tau(\omega) > 0 \quad \text{for all } \omega.$$

Stable & causal All Pass systems have Positive group delay for all ω.

1.a Cascading of many all pass systems leads to an all pass system

$$H(z) = \prod_{k=1}^{N} \frac{z^{-1} - p_k^*}{1 - p_k z^{-1}}$$

$p_1, p_2, \ldots, p_N$ are poles

$\frac{1}{p_1^*}, \frac{1}{p_2^*}, \ldots, \frac{1}{p_N^*}$ are zeros

$$H(z) = \frac{z^{-N} + a_1^* z^{-(N-1)} + \ldots + a_{N-1}^* z^{-1} + a_N^*}{a_N z^{-N} + \ldots + a_1 z^{-1} + 1}$$

N^{th} order all pass system

All pass systems are used in phase compensation system.

2. Minimum Phase systems

(Minimum group delay systems)

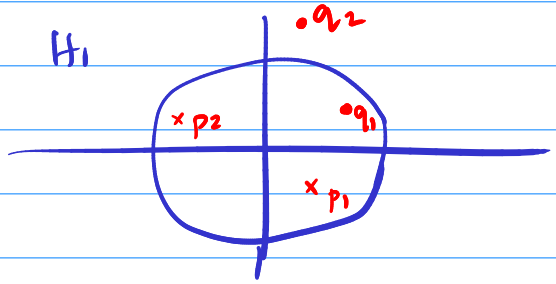
(Among all systems with same magnitude response)

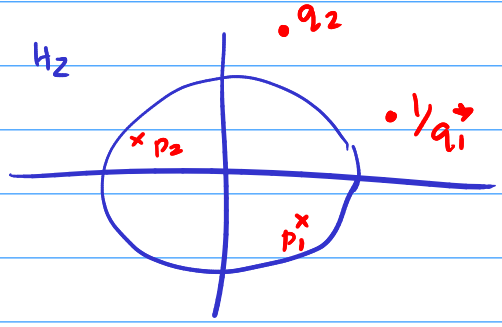
Example

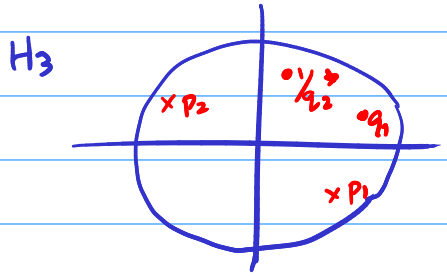
$$H_1(z) = \frac{(1-q_1 z^{-1})(1-q_2 z^{-1})}{(1-p_1 z^{-1})(1-p_2 z^{-1})}$$

Poles @ p_1, p_2

Zeros @ q_1, q_2







There are 16 such systems with same mag. resp as H_1.

$H_3(z)$ has all the poles and zeros inside unit circle.

Definition A stable and causal system with all the poles and zeros inside unit circle is called Minimum phase system.

Remark. $H(z)$ minimum phase system

$G(z)$ Inverse of min. phase system

$$G(z) = \frac{1}{H(z)}$$

Poles of G = Zeros of H

Zeros of G = Poles of H

Inverse G is also a min. phase system

It is also stable